

Anthony (Yu-Chung) Cheng

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Education

Virginia Tech, *Master of Engineering in Computer Science*

Dec 2024

GPA: **3.97/4.0**

Blacksburg, VA

Research: CreativeLab: Unique Concept Generation Using Image Clustering ([Research Report](#))

National Tsing Hua University, *Master of Science in Computer Science*

Jan 2017

GPA: **3.9/4.3**

Hsinchu, Taiwan

Thesis: *Performance Improvement of OpenFlow Switch with Queue and Per-port Cache* (Thesis)

Research Experience

Creative Image Generation with Unsupervised Clustering (PyTorch)

2024

- Built clustering-based pipelines for novel concept generation; focused on comparative learning and loss design.
- Code: [ConceptLab \(cluster branch\)](#).
- Acknowledgements to Prof. Pinar Yanardag; no conference submission.

Using GAN to Enhance Multi-Object Tracking (FairMOT)

Fall 2023

- Integrated a GAN discriminator with a pre-trained **FairMOT** model to improve tracking robustness.
- Code: github.com/sunnyanthony/cv-project-fall23.

Data Augmentation for Image Segmentation via Diffusion Models

Spring 2023

- Implemented diffusion-based augmentation to improve segmentation accuracy and generalization.
- Notebook: [diffusion_playground \(aml_project\)](#).

Master's Thesis: Performance Improvement of OpenFlow Switch with Queue and Per-port Cache (NTHU) 2014–2016

- Investigated queue and per-port cache strategies to improve OpenFlow switch efficiency and performance in SDN.
- Conducted under Prof. Youn-Long Lin; thesis: [link](#).

Selected Projects

Computer Vision Projects

2023

- **Image Stitching & Epipolar Geometry**: feature matching, homography estimation, epipolar constraints.
- **YOLO customizations**: trained/tuned object detectors; evaluated accuracy/latency trade-offs.
- **Adversarial Attack Defenses**: attack/defense techniques and measured robustness on object detection model.
- **Image Generation with LSGAN & BEGAN**: reproduced and compared generative behaviors and stability.
- **CLIP few-shot classification**: Tip-Adapter on mini-ImageNet; explored **CoOp** for prompt learning.

C++ Relational Database (B+tree, SQL Executor, Concurrency)

(Carnegie Mellon University)

- Implemented storage (B+tree), query execution, and concurrency control in; achieved **top 10%** performance.

ETDs Information Retrieval System & LLM Recommender on Kubernetes (Rancher)

Fall 2023

- Coordinated infra across teams; engineered 20+ microservices with **Kafka**, **Elasticsearch**, **PostgreSQL**, **LLaMA2**, and **Flask** for a RAG pipeline; automated **GitLab CI/CD**.
- Enforced service-token authentication and request verification; course site: cs5604.endeavour.cs.vt.edu.

Developing Backend for OpenAI Applications (FastAPI)

Spring 2024

- Built backend integrations with **ChatGPT** and **DALL-E**.
- Implemented a multi-agent code vulnerability checker producing reports and attack code (for testing).

Big Data Processing on AWS with Spark (EMR + S3)

2023

- Built an **EMR** service to process data from **S3** with **Spark** for cleaning and aggregation.

Publications and Patents

System and method for reducing surveillance detection errors, Published: [US20230095027A1](#) (Mar 2023)

Teaching Experience

Teaching Assistant, National Tsing Hua University

Jan 2015 – Jun 2016

- Led tutorials and lab sessions for Algorithms and Digital Logic Design; assisted grading and student mentoring.

Professional Experience

Amazon.com, Washington, USA

Jan 2025 —

Software Development Engineer II

AWS, SageMaker HyperPod

- Developed scalable **ML infrastructure** for **distributed training** of foundation models; delivered customized training environments (e.g., Nova/Amazon Q) to reduce setup time and improve reliability.

Software Development Engineer (L5)

AFT, Identity

- Developed and maintained a global **authN/authZ** platform supporting large-scale daily active users.

Deep Sentinel (AI Startup, Series A), Hybrid

Mar 2021 – Nov 2022

Senior Software Engineer

- Architected distributed microservices on **GCP** with message queuing and monitoring to optimize response times.
- Optimized **WebRTC** reliability (**GStreamer** + signaling), maintaining **99%** low-latency live streaming stability.
- Enhanced database performance via **Redis** caching/partitioning/indexing, reducing peak query time by **75%**.
- Built a distributed-cache-based **workload balancer** for event-driven systems, achieving **90%** efficiency.
- Collaborated on an ML similarity-check pipeline, reaching up to **F1 80%**.
- Designed a hybrid-cloud protocol for **SQLite/MySQL**, reducing data delay by **35%** and improving availability.

Realtek, Taipei, Taiwan

Apr 2017 – Mar 2021

Senior Software Engineer / Software Engineer

- Built embedded networking SDK, toolchains, and Linux/FreeRTOS applications with full test coverage and documentation.
- Integrated **MbedTLS** and a security accelerator to harden open-source stacks; reduced cipher time by **70%**.
- Implemented virtualization-based **zero-copy** to improve Linux/FreeRTOS throughput by **150%**.
- Designed a key-value lightweight DB cutting query time by **50%**
- Innovated backtracking algorithm in the linker to **60%** reduce memory fragmentation.
- Drove **CI/CD** (Jenkins/GitLab), improving team productivity by **30%**.

Technical Skills

Programming Languages: Python, C++, C, Rust, Java

ML/AI: PyTorch, Sklearn, OpenCV; Diffusion, GANs

Systems/Parallel: Distributed Systems, Concurrency; CUDA, OpenMP, MPI

Data/DB: PostgreSQL, MySQL, Redis; B+tree

Cloud/Infra: AWS, GCP, Docker, Kubernetes

Security: MbedTLS, OAuth2, OIDC

Certifications

Cryptography (Stanford)

Machine Learning (Stanford)